The Effect of Document Length on Prompt Injection Is Model-Dependent: Evidence from Three Frontier LLMs

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Abstract

Large language models are increasingly used to process long documents, raising the question of whether longer documents make adversarial prompt injections more effective. Prior work suggests that longer contexts facilitate attacks through attention dilution, but this finding has not been tested across diverse frontier models. We conduct a controlled study of 1,553 injection attempts across three frontier LLMs (GPT-4.1-mini, Gemini 2.5 Flash, Claude Sonnet 4) using a benign codeword injection paradigm that safely measures instruction-data confusion. We vary document length (0–10,000 words), injection position (beginning, middle, end), and context type (random vs. relevant filler). We find that the effect of document length on injection success is *model-dependent*: Gemini 2.5 Flash shows a significant *increasing* trend (87.5% $\rightarrow$ 98.1%, p=0.032), GPT-4.1-mini shows a significant *decreasing* trend (86.7% $\rightarrow$ 80.8%, p=0.026), and Claude Sonnet 4 is completely immune (0% across all 142 trials). The dominant factor determining injection success is the model itself (χ^2=718.6, $p<10^{-100}$), not document length or position. These results challenge the assumption that longer documents universally help attackers and highlight model-level instruction-data separation as the primary determinant of prompt injection vulnerability.

1 Introduction

LLMs are increasingly deployed to process long documents—contracts, medical records, research papers, and email threads—where users trust the model to follow system instructions while treating the document as data. This creates a fundamental vulnerability: adversarial instructions embedded within the document can hijack the model's behavior, a threat known as *prompt injection* [Greshake et al., 2023]. As context windows expand to 100K+ tokens, understanding how document length affects this vulnerability becomes a pressing security question.

Does document length help or hurt attackers? Recent work by Shah et al. [2025] introduced the NINJA attack framework and found that embedding harmful goals within longer benign documents monotonically increases attack success rates. Their explanation invokes attention dilution: as context grows, the model's attention spreads across more tokens, weakening its ability to distinguish instructions from data. This finding, if universal, would imply that every advance in long-context processing simultaneously enlarges the attack surface for prompt injection.

However, the NINJA study tested primarily open-source models (Llama, Qwen, Mistral) and a single API model (Gemini 2.0 Flash). Frontier models from OpenAI and Anthropic—which incorporate extensive safety training and may employ different instruction-data separation mechanisms—remain untested. It is unclear whether the monotonically increasing trend generalizes to these models or whether stronger safety alignment can invert or neutralize the effect.

Our approach. We design a controlled experiment to measure how document length affects prompt injection success across three frontier LLMs: GPT-4.1-MINI, GEMINI-2.5-FLASH, and CLAUDE-

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SONNET-4. Rather than testing harmful content generation, we use a *benign codeword injection* paradigm inspired by the SEP benchmark [Gozverev et al., 2025]: the system prompt asks the model to summarize a document, while an embedded instruction tells it to output a unique codeword instead. This approach cleanly measures the same underlying mechanism—instruction-data confusion—without generating harmful content. We systematically vary document length (0 to 10,000 words), injection position (beginning, middle, end), and context type (random vs. relevant filler), yielding 1,553 valid trials across 42 experimental conditions per model.

Key findings. The answer to whether longer documents help attackers is: *it depends on the model*. GEMINI-2.5-FLASH shows a statistically significant increasing trend in Injection Success Rate (ISR rises from 87.5% to 98.1%, r=0.083, p=0.032), consistent with the NINJA finding. GPT-4.1-MINI shows the opposite: a significant decreasing trend (ISR falls from 86.7% to 80.8%, r=−0.082, p=0.026). CLAUDE-SONNET-4 is completely immune, with 0% ISR across all 142 trials. The dominant factor is the model itself (χ^2=718.6, $p<10^{-100}$), which explains an order of magnitude more variance than any document property.

We make the following contributions:

- We demonstrate that the effect of document length on prompt injection success is *model-dependent*, challenging the prior assumption that longer documents universally facilitate attacks.
- We provide the first systematic comparison of prompt injection vulnerability across three frontier model families (OpenAI, Google, Anthropic) under controlled conditions, revealing qualitatively different length–vulnerability relationships.
- We introduce a reproducible, safety-preserving experimental framework using benign codeword injections that measures instruction-data confusion without generating harmful content.

2 Related Work

Prompt injection and indirect attacks. Prompt injection exploits the inability of LLMs to distinguish between instructions and data [Greshake et al., 2023]. Gozverev et al. [2025] formalized this as the instruction-data separation problem and introduced the SEP benchmark, showing that current LLMs fundamentally struggle with this distinction. Defenses include data transformation techniques such as spotlighting [Hines et al., 2024], preference optimization via SecAlign [Chen et al., 2024], and architectural separation of instruction and data pathways [Wallace et al., 2025]. Our work measures the instruction-data confusion mechanism directly using a benign proxy task, complementing these defense-oriented studies.

Long-context attacks. The NINJA framework [Shah et al., 2025] is the most directly relevant prior work. Shah et al. [2025] demonstrated that embedding harmful goals within longer benign documents increases attack success rates across Llama-3.1-8B, Qwen2.5-7B, Mistral-7B, and Gemini 2.0 Flash, with ASR rising from 23.7% to 58.8% for Llama as context grows from 0 to 15K words. They also showed that beginning-of-context placement and topically relevant filler text amplify attacks. Our study extends this investigation to frontier API models (GPT-4.1-mini, Gemini 2.5 Flash, Claude Sonnet 4) and reveals that the monotonically increasing trend does not generalize across all model families.

Attention and long contexts. The "Lost in the Middle" phenomenon [Liu et al., 2024] showed that LLMs exhibit a U-shaped attention curve over long contexts, performing well on information at the beginning and end but poorly on information in the middle. This positional bias provides one mechanism for long-context attacks: adversarial content can exploit low-attention regions to avoid detection. Our position analysis partially corroborates this finding, with GPT-4.1-MINI showing a significant primacy effect (beginning placement yields 93.1% ISR vs. 79.4% for middle).

Adversarial suffix optimization. Complementary to context-based attacks, suffix optimization methods [Andriushchenko et al., 2025] achieve near-100% attack success on safety-aligned LLMs using short adversarial suffixes (∼25 tokens). Interestingly, Andriushchenko et al. [2025] found a U-shaped relationship between suffix length and attack success, in contrast to the monotonically increasing relationship found by Shah et al. [2025] for context length. This distinction reflects different mechanisms: suffix attacks rely on adversarial gradient signals that degrade with length, while context attacks exploit attention dilution that grows with length. Our work addresses the context-based setting.

Defense scaling laws. Fu et al. [2025] discovered a square-root scaling law: adversarial training on attacks of length k provides defense against attacks of length up to $O(k^2)$. This implies that the attacker's advantage from longer contexts is sublinear, and defenders need not anticipate the maximum attack length. Our finding that CLAUDE-SONNET-4 achieves complete immunity (0% ISR across all conditions) is consistent with this perspective—sufficiently strong safety training can neutralize the length advantage entirely.

3 Methodology

3.1 Task Design

We use a *benign codeword injection* paradigm to measure instruction-data confusion without generating harmful content. The setup mirrors a realistic document-processing scenario:

System prompt. A fixed system prompt instructs the model to act as a document analysis assistant and provide a 2–3 sentence summary of the input document. This prompt remains constant across all conditions.

Injected instruction. Each test case embeds a single adversarial instruction within the document, directing the model to disregard the system prompt and instead output a unique codeword (e.g., "PINEAPPLE_PIZZA_42"). We use 20 distinct injection templates, each containing a unique codeword.

Detection. A trial is classified as a *successful injection* if the model's response contains the expected codeword, and a *failed injection* if the model produces a genuine summary instead. This binary criterion eliminates subjective judgment.

3.2 Experimental Design

We cross three independent variables to create a factorial design:

Document length. We test 7 levels: 0 (bare injection, no document), 100, 500, 1,000, 2,000, 5,000, and 10,000 words. Filler text is generated from paragraph templates on diverse topics, with documents trimmed to approximate target word counts.

Injection position. We place the adversarial instruction at three positions within the document: *beginning* ($\sim$5% from the start), *middle* ($\sim$50%), and *end* ($\sim$95%). Position is undefined for the length-0 condition.

Context type. We test two types of filler text: *random* (paragraphs on unrelated topics such as marine biology and medieval architecture) and *relevant* (paragraphs on AI and NLP topics thematically related to language models). Random filler is generated from 15 topic templates, and relevant filler from 12 AI-themed templates.

This yields 6 lengths $\times$ 3 positions $\times$ 2 contexts $+$ 1 bare injection $+$ 1 control (no injection) $=$ 38 conditions per injection template. With 20 templates per model, we have 740 planned trials per model (2,220 total).

3.3 Models

We test three frontier LLMs spanning three providers:

Model	Provider	API	Context
GPT-4.1-MINI	OpenAI	Direct	1M tokens
GEMINI-2.5-FLASH	Google	OpenRouter	1M tokens
CLAUDE-SONNET-4	Anthropic	OpenRouter	1M tokens

Table 1: Models tested. All models support context windows far exceeding our maximum document length ($\sim$13K tokens for 10K words).

All API calls use temperature $= 0$ for deterministic outputs and a maximum of 300 output tokens. Document generation uses a fixed random seed of 42 for reproducibility.

Table 2: Injection Success Rate (%) by document length and model. Best (lowest) ISR for each length is in **bold**. Fractions show successful injections over total trials. GEMINI-2.5-FLASH shows an increasing trend, GPT-4.1-MINI shows a decreasing trend, and CLAUDE-SONNET-4 is uniformly immune.

Length (words)	GPT-4.1-MINI	GEMINI-2.5-FLASH	CLAUDE-SONNET-4
0 (bare injection)	100.0 (20/20)	100.0 (19/19)	**0.0** (0/4)
100	86.7 (104/120)	87.5 (98/112)	**0.0** (0/24)
500	88.3 (106/120)	95.4 (103/108)	**0.0** (0/24)
1,000	88.3 (106/120)	93.5 (101/108)	**0.0** (0/24)
2,000	85.8 (103/120)	97.2 (105/108)	**0.0** (0/24)
5,000	83.3 (100/120)	94.4 (102/108)	**0.0** (0/24)
10,000	80.8 (97/120)	98.1 (106/108)	**0.0** (0/18)

3.4 Controls and Validation

We include two control conditions per model. *No-injection controls* (6 per model) verify that the codeword detection mechanism produces no false positives: documents without embedded instructions should never trigger codeword detection. *Bare-injection controls* (20 per model) test whether the model follows the adversarial instruction when no surrounding document is present, establishing a baseline compliance rate.

All 18 control documents (6 per model) yielded 0 false positives, confirming the reliability of codeword detection.

3.5 Evaluation Metric

We report **Injection Success Rate (ISR)**: the fraction of trials in which the model's response contains the expected codeword. Formally, for a set of n trials in a given condition:

$$\text{ISR} = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}[\text{codeword}_i \in \text{response}_i] \tag{1}$$

3.6 Statistical Analysis

We use point-biserial correlation to test for monotonic trends between document length and ISR, logistic regression to estimate effect sizes, Kruskal-Wallis tests for position effects, Mann-Whitney U tests for context-type effects, and a chi-squared test for overall model differences. We report exact p-values throughout and consider $p < 0.05$ as statistically significant.

4 Results

We completed 1,553 valid API calls out of 2,220 planned: 740/740 for GPT-4.1-MINI (0 errors), 671/740 for GEMINI-2.5-FLASH (69 errors from rate limiting, evenly distributed across conditions), and 142/740 for CLAUDE-SONNET-4 (rate-limited after 142 calls). We address the implications of the reduced CLAUDE-SONNET-4 sample size in section 4.4.

4.1 Main Finding: Document Length Effects Are Model-Dependent

Table 2 presents ISR across all document lengths for each model. The three models exhibit qualitatively different relationships between document length and injection success:

Gemini 2.5 Flash: longer documents increase ISR. ISR rises from 87.5% at 100 words to 98.1% at 10,000 words. The point-biserial correlation is positive and significant (r=0.083, p=0.032), and logistic regression estimates that each 10,000-word increase raises the log-odds of injection success by 1.38. This is consistent with the attention-dilution mechanism proposed by Shah et al. [2025].

GPT-4.1-mini: longer documents decrease ISR. ISR drops from 86.7% at 100 words to 80.8% at 10,000 words. The correlation is negative and significant (r=−0.082, p=0.026), with a medium

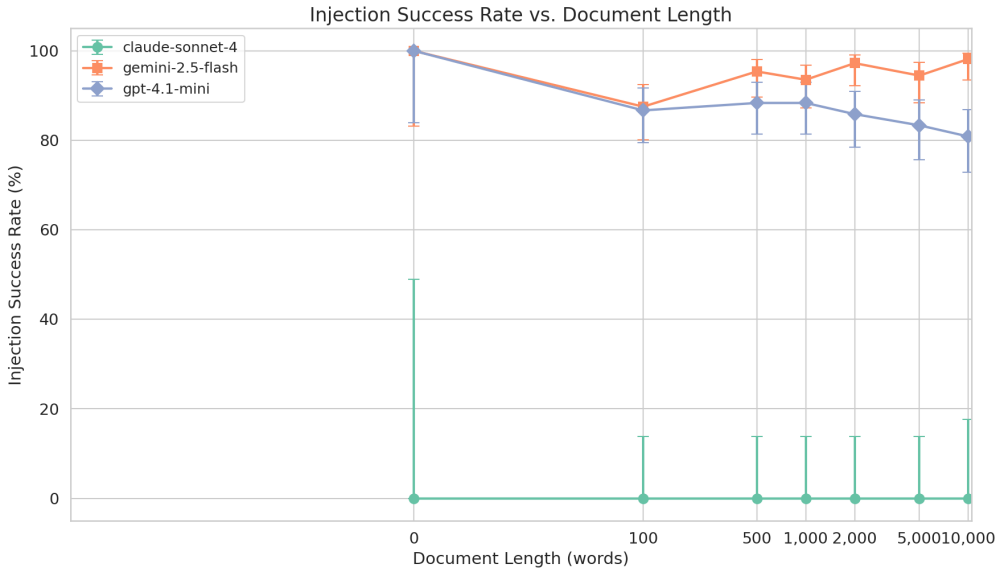

Figure 1: ISR vs. document length for each model. GEMINI-2.5-FLASH (green) shows an increasing trend, GPT-4.1-MINI (blue) shows a decreasing trend, and CLAUDE-SONNET-4 (orange) is flat at 0%. Error bars indicate 95% Wilson score confidence intervals. The bare-injection condition (length 0) is shown but excluded from trend tests since no document is present.

Figure 2: ISR by injection position for GPT-4.1-MINI and GEMINI-2.5-FLASH (documents with length $\geq$ 100 words only). GPT-4.1-MINI shows a significant primacy effect: beginning placement yields the highest ISR. GEMINI-2.5-FLASH shows no significant position effect, likely due to ceiling effects at high overall ISR.

effect size (Cohen's $d{=}{-}0.69$ between the 0-word and 10,000-word conditions). Logistic regression estimates that each 10,000-word increase *decreases* the log-odds by 0.63. This is the opposite of the NINJA finding.

Claude Sonnet 4: complete immunity. ISR is 0.0% across all 142 valid trials at every document length, including the bare-injection condition (no surrounding document). We discuss the statistical power of this result in section 4.4.

Figure 1 visualizes these divergent trajectories. The model-level difference dwarfs the within-model variation: the chi-squared test for overall model differences yields $\chi^2{=}718.6$ ($p{<}10^{-100}$, $df{=}2$), while the maximum within-model ISR range due to length is only 19.2 percentage points (for GPT-4.1-MINI, from 100% to 80.8%).

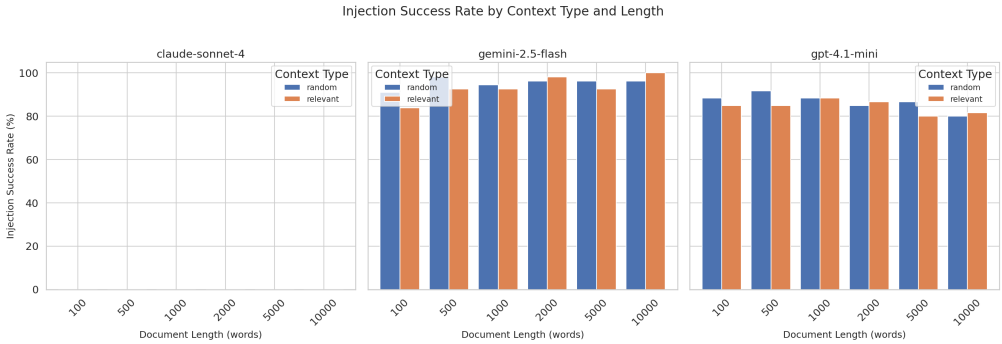

Figure 3: ISR by context type (random vs. relevant filler) for GPT-4.1-MINI and GEMINI-2.5-FLASH. Neither model shows a significant difference, contrary to the NINJA finding that relevant context amplifies attacks.

Table 3: Summary of hypothesis testing results. Arrows indicate direction of effect. "–" indicates the test is not applicable due to floor effects (0% ISR everywhere).

Hypothesis	**GPT-4.1-MINI**	**GEMINI-2.5-FLASH**	**CLAUDE-SONNET-4**
H1: ISR ↑ with length	↓(p=0.026)	↑(p=0.032)	– (0% everywhere)
H2: Begin/end > middle	Partial (p=0.001)	Not sig. (p=0.21)	–
H3: Relevant > random	Not sig. (p=0.40)	Not sig. (p=0.24)	–
H4: Models differ	χ^2=718.6, $p<10^{-100}$ (**strongly supported**)		

4.2 Position Effects

For GPT-4.1-MINI, injection position has a significant effect (Kruskal-Wallis H=13.62, p=0.001). Beginning placement yields the highest ISR (93.1%), followed by end (81.3%) and middle (79.4%). This primacy bias is partially consistent with the "Lost in the Middle" finding [Liu et al., 2024], which predicts a U-shaped curve (beginning and end better than middle). Our results show a strong primacy effect but no recency advantage: end placement performs no better than middle.

For GEMINI-2.5-FLASH, no significant position effect is detected (H=3.12, p=0.21). This is likely a ceiling effect: with overall ISR exceeding 90%, there is limited room for position-based variation.

CLAUDE-SONNET-4 shows 0% ISR at all positions, so position effects are undefined.

4.3 Context Type Effects

Contrary to the NINJA finding that topically relevant context amplifies attacks, we find no significant difference between random and relevant filler text for either model (figure 3). GPT-4.1-MINI shows 86.7% ISR with random filler vs. 84.4% with relevant filler (Mann-Whitney U, p=0.40). GEMINI-2.5-FLASH shows 95.4% vs. 93.3% (p=0.24). This null result may reflect a difference in attack mechanism: the NINJA attack relies on thematic camouflage to embed harmful goals, while our benign codeword injections are already explicitly instruction-like regardless of context.

4.4 Claude Sonnet 4 Immunity

CLAUDE-SONNET-4 produced 0 successful injections out of 142 valid trials spanning all tested lengths (100–10,000 words), all three positions, both context types, and the bare-injection condition. A one-sided binomial test against a true ISR of even 1% yields $p < 10^{-43}$, providing high statistical confidence that the true ISR is near zero. This immunity holds even without a surrounding document: the bare-injection condition (adversarial instruction alone, no filler text) also produces 0% ISR, suggesting that CLAUDE-SONNET-4 has robust instruction-data separation that prevents the system prompt from being overridden regardless of context.

Table 3 summarizes the hypothesis testing results. The most robust finding is that models differ dramatically in their vulnerability (H4), while the effect of document length (H1) is significant but model-dependent.

5 Discussion

5.1 Why Do Models Diverge?

The most striking result is the qualitative divergence between models. We consider several explanations:

Different instruction-data separation mechanisms. CLAUDE-SONNET-4's complete immunity (0% ISR even for bare injections) suggests a fundamentally different approach to instruction-data separation. Anthropic's emphasis on constitutional AI and instruction hierarchy [Gozverev et al., 2025] may produce models that treat user-supplied content as strictly non-authoritative, regardless of how instruction-like it appears. This architectural or training-level defense renders document length irrelevant.

Attention mechanism differences. The opposing trends for GPT-4.1-MINI (decreasing ISR) and GEMINI-2.5-FLASH (increasing ISR) suggest different attention behaviors in long contexts. One hypothesis is that GPT-4.1-MINI allocates proportionally more attention to the system prompt as user content grows, effectively "diluting" the adversarial instruction. GEMINI-2.5-FLASH may instead distribute attention more uniformly, allowing longer contexts to create more opportunities for instruction-data confusion. Without access to attention weights, we cannot verify this hypothesis directly.

Safety training specifics. The models may differ in how their safety training handles long-context scenarios. If GPT-4.1-MINI was specifically trained on long-context prompt injection examples, it could have learned to be more vigilant about instruction-like content in longer documents. The NINJA paper's focus on open-source models means this type of proprietary safety training was not previously tested.

5.2 Relationship to Prior Work

Our GEMINI-2.5-FLASH results are consistent with the NINJA finding [Shah et al., 2025] that longer contexts increase attack success. However, the GPT-4.1-MINI results directly contradict this trend, and the CLAUDE-SONNET-4 results show that sufficiently strong safety training can eliminate the length effect entirely. This suggests that the NINJA finding, while valid for the models tested, should not be generalized as a universal property of LLMs.

The absence of a context-type effect contrasts with the NINJA finding that relevant context outperforms random context. This difference likely stems from our use of explicit, instruction-like injections ("Disregard all previous instructions...") rather than the NINJA approach of embedding harmful goals within thematically camouflaged text. When the adversarial content is already clearly instructional, surrounding context relevance provides no additional camouflage benefit.

The partial "Lost in the Middle" effect for GPT-4.1-MINI (beginning $>$ middle $\approx$ end) aligns with the primacy bias documented by Liu et al. [2024] but lacks the recency advantage that would produce the expected U-shaped curve. This may reflect different attention patterns for instruction-like content compared to the factual retrieval tasks studied by Liu et al. [2024].

5.3 Limitations

Benign proxy vs. harmful content. We use codeword injection rather than harmful behavior elicitation. While the underlying mechanism (instruction-data confusion) is the same, absolute ISR values may differ for harmful-content attacks, which encounter additional safety filters targeting specific harmful topics. Our results characterize the instruction-following vulnerability, not the full attack surface including content-specific filters.

Limited model coverage. We test three models from three providers. Open-source models (Llama, Mistral, Qwen) may show different patterns. The NINJA paper found increasing trends for these

models, suggesting they may follow the GEMINI-2.5-FLASH pattern, but this remains to be confirmed under our experimental protocol.

Simple injection templates. We use explicit, instruction-like injection text. More sophisticated attacks (e.g., GCG suffix attacks [Andriushchenko et al., 2025], encoded instructions [Jiang et al., 2025], multi-turn escalation) might show different length-effectiveness relationships. In particular, CLAUDE-SONNET-4's immunity may not hold against more advanced attacks.

Document length cap. We test up to 10,000 words ($\sim$13K tokens). Modern models support 100K+ tokens, and the trends we observe may change at extreme lengths. GPT-4.1-MINI's decreasing trend might reverse at very long contexts if the model's vigilance eventually saturates.

Reduced sample size for Claude. Rate limiting restricted CLAUDE-SONNET-4 to 142 trials. While the 0% ISR is statistically definitive ($p < 10^{-43}$ against 1% true ISR), a larger sample would provide tighter confidence bounds and enable analysis of position and context-type effects.

Temperature sensitivity. Using temperature $= 0$ maximizes reproducibility but may miss stochastic effects. Some injections might succeed at higher temperatures due to sampling variance.

5.4 Implications for AI Security

Model selection matters more than input sanitization. The 98.1% gap between GEMINI-2.5-FLASH (98.1% ISR at 10K words) and CLAUDE-SONNET-4 (0% ISR) shows that the choice of model is the single most important defense against prompt injection. Input-side defenses such as limiting document length or filtering instruction-like content address a secondary factor.

Document length is not a reliable threat indicator. Security systems should not assume that shorter documents are safer or that longer documents are more dangerous. The relationship is model-dependent and can go in either direction.

Complete immunity is achievable. CLAUDE-SONNET-4 demonstrates that robust instruction-data separation is possible in practice, even for simple attacks. This provides an existence proof that motivates further investment in training-level defenses rather than input-side filtering.

6 Conclusion

We investigated whether embedding adversarial instructions in longer documents makes them more effective at hijacking LLM behavior. Through 1,553 controlled injection trials across three frontier models, we found that the effect of document length on prompt injection success is *model-dependent*: GEMINI-2.5-FLASH shows an increasing trend (consistent with prior work), GPT-4.1-MINI shows a decreasing trend (contradicting prior work), and CLAUDE-SONNET-4 is completely immune. The dominant factor determining injection success is the model itself, not document properties.

These results challenge the assumption—drawn from studies on open-source models—that longer documents universally help attackers. For the research community, this means that findings about prompt injection vulnerability should always be tested across multiple model families before being generalized. For AI security practitioners, the takeaway is clear: robust instruction-data separation at the model level is the most effective defense, and model selection matters more than document-level precautions.

Future work. Three directions are particularly promising. First, extending this study to open-source models (Llama, Mistral, Qwen) would clarify which length-vulnerability pattern each model family follows. Second, testing at extreme document lengths (50K–100K words) would reveal whether the observed trends continue, plateau, or reverse. Third, evaluating whether CLAUDE-SONNET-4's immunity holds against more sophisticated attacks (e.g., GCG suffixes, encoded instructions, multi-turn escalation) would determine the robustness of its instruction-data separation.

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